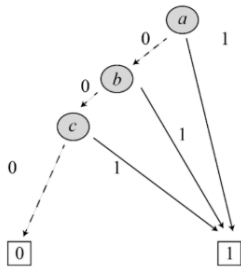


1. Consider the following ROBDD:

1 Point



Which of the following functions will have the above ROBDD?

- A. $a+b+c$
- B. $a+b+b'c$
- C. $a+a'b+a'c$

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☐ only A, C
- ☐ only B, C
- ☒ All
- ☐ None

Yes, the answer is correct. Score: 1

Accepted answer is All

2. Consider the following function:

1 Point

$$y = ab + acd + b'd + bc'$$

Consider the following statements:

- A. The positive cofactor of the above function with respect to the variable a is $b + cd + b'd + bc'$
- B. The positive cofactor of the above function with respect to the variable b is $b + cd + b'd + bc'$
- C. The positive cofactor of the above function with respect to the variable c is $ab + ad + b'd$

Which of the above statements are correct?

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☒ only A, C
- ☐ only B, C
- ☐ All statements are correct.
- ☐ No statement is correct.

Yes, the answer is correct. Score: 1

Accepted answer is only A, C

3. Consider the following statements about the ROBDD of a Boolean function:

1 Point

- A. The number of vertices in an ROBDD always increases linearly with the number of variables.
- B. The size of the ROBDD depends on the variable order chosen in the representation.
- C. It is a canonical representation of a Boolean function.

Which of the statements are correct?

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☐ only A, C
- ☒ only B, C
- ☐ All statements are correct.
- ☐ No statement is correct.

Yes, the answer is correct. Score: 1

Accepted answer is only B, C

4. Consider the following statements regarding functional verification:

1 Point

- A. We typically need to provide an exhaustive set of test stimuli to a formal verification tool, which then simulates and checks using those stimuli.
- B. Among the given set of stimuli, a formal verification tool randomly chooses a subset of stimuli for which the simulation is done.
- C. Formal verification tools typically employ ROBDD and SAT solvers to establish proof of correctness.

Which of the statements are correct?

- ☐ only A
- ☐ only B
- ☒ only C
- ☐ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All statements are correct.
- ☐ No statement is correct.

Yes, the answer is correct. Score: 1

Accepted answer is only C

5. Consider the following statements regarding finite state machines:

1 Point

- A. Given the current state and the current input, the state transition function produces the next state.
- B. The output of the FSM is a function of the current state only in a Mealy Machine.
- C. The output of the FSM is a function of the current state and current input in a Moore Machine.

Which of the statements are correct?

- ☒ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All statements are correct.
- ☐ No statement is correct.

Yes, the answer is correct. Score: 1

Accepted answer is only A

6. Which of the following techniques are used in FSM optimization?

1 Point

- A.State minimization
- B.State encoding that minimizes the hardware cost
- C.Transforming the state of matter

- ☐ only A
- ☐ only B
- ☐ only C
- ☒ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All
- ☐ None

Yes, the answer is correct. Score: 1

Accepted answer is only A, B

7. To represent an FSM with n_s states, the minimum number of bits needed is:

1 Point

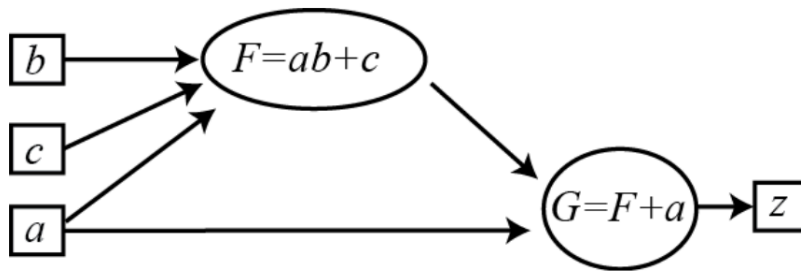
- ☒ $\lceil \log_2 n_s \rceil$
- ☐ $\lceil \log_{10} n_s \rceil$
- ☐ n_s
- ☐ $n_s + 1$

Yes, the answer is correct. Score: 1

Accepted answer is $\lceil \log_2 n_s \rceil$

8. Which of the following expressions represent observability don't care (ODC) for F in the following Boolean logic network?

1 Point



- ☐ a'
- ☒ a
- ☐ b'
- ☐ b
- ☐ c'
- ☐ c

Yes, the answer is correct. Score: 1

Accepted answer is a

9. Match the following transformations done during multi-level logic optimization with their descriptions:

1 Point

Transformations	Description
A. Eliminate	1. Two-level logic optimization is carried out on each local function individually.
B. Simplify	2. Finds a common subexpression for functions associated with two or more vertices.
C. Substitute	3. Removes a vertex from the graph and replaces all its occurrences in the network with the corresponding local function.
D. Extract	4. Replaces the local function with a simpler SOP by creating new dependencies and possibly removing other dependencies.

- ☐ A-1, B-2, C-3, D-4
- ☐ A-1, B-3, C-4, D-1
- ☐ A-2, B-1, C-4, D-3
- ☐ A-2, B-4, C-1, D-3
- ☒ A-3, B-1, C-4, D-2
- ☐ A-3, B-4, C-1, D-2
- ☐ None of the above

Yes, the answer is correct. Score: 1

Accepted answer is A-3, B-1, C-4, D-2

10. Which of the following tools can be used for RTL synthesis?

1 Point

- ☐ OpenSTA
- ☒ Yosys
- ☐ Tempus
- ☐ PrimeTime

Yes, the answer is correct. Score: 1

Accepted answer is Yosys